

- $\delta_{B1} = 1$ , if the Birmingham repair capacity is expanded by 5 cars per day.  
 $\delta_{B2} = 1$ , if the Birmingham repair capacity is expanded by a further 5 cars per day.  
 $\delta_{M1} = 1$ , if the Manchester repair capacity is expanded by 5 cars per day.  
 $\delta_{M2} = 1$ , if the Manchester repair capacity is expanded by a further 5 cars per day.  
 $\delta_P = 1$ , if the Plymouth repair capacity is expanded by 5 cars per day.

The following expression is added to the objective function:

$$18\,000\delta_{B1} + 8\,000\delta_{B2} + 20\,000\delta_{M1} + 5\,000\delta_{M2} + 19\,000\delta_P$$

together with the following extra constraints:

$$\delta_{B1} \geq \delta_{B2}, \quad \delta_{M1} \geq \delta_{M2}, \quad \delta_{B1} + \delta_{B2} + \delta_{M1} + \delta_{M2} + \delta_P \leq 3$$

and the repair capacity constraints for Birmingham, Manchester and Plymouth have  $5\delta_{B1} + 5\delta_{B2}$ ,  $5\delta_{M1} + 5\delta_{M2}$  and  $5\delta_P$ , respectively added.

## 13.27 Lost baggage distribution

We formulate this problem as an integer programming model. It is an extension of the travelling salesman problem, discussed in section 9.5. It can be regarded as a (simple) case of the vehicle routing problem. In the problem here, there are no vehicle capacities and no time windows for the delivery to each location (apart from the overall 2 hours guarantee). Nevertheless, it is potentially a very difficult problem to solve, for more than a modest number of locations. Also it can give rise to a very large model. There are a number of different ways of formulating a model, which vary greatly in their size and difficulty of solution. We suggest a model that is practicable in both size and computational tractability. Although the problem is 'symmetric' in the sense that the distance from  $X$  to  $Y$  is the same as that from  $Y$  to  $X$ , we extend the asymmetric formulation of the travelling salesman problem in order to give the 'time' to return to Heathrow, after a van has completed all its deliveries, as 0. In this way, only the total time taken to reach the last delivery is counted within the 2 hours time limit. We are therefore seeking a 'Hamiltonian path' (as opposed to a circuit) for each van, starting from Heathrow.

### 13.27.1 Variables

All variables are 0–1 and integer

$x_{ijk} = 1$ , iff van  $k$  visits, and goes directly from location  $i$  to  $j$  for all  $i, j$ , and  $k$ .

$y_{ik} = 1$ , iff location  $i$  is visited by van  $k$  for all  $i, k$ .

$\delta_k = 1$ , iff van  $k$  is used.

### 13.27.2 Objective

Minimize  $\sum_k \delta_k$ , that is, minimize the number of vans used.

### 13.27.3 Constraints

$y_{ik} \leq \delta_k$  for all  $i > 1, j, k$  if a location is visited by van  $k$ , then it is (obviously) used

$\sum_{ij: i < j} c_{ij} x_{ijk} + \sum_{ij: i > j: j \neq 1} c_{ji} x_{jik} \leq 120$  for all  $k$ , no van travels for more than 2 hours. The time from any location back to Heathrow does not count

$\sum_k y_{ik} = 1$  for all  $i$  (apart from  $i = 1 = \text{Heathrow}$ ), i.e. each location is visited by exactly one van

$\sum_k y_{1k} \geq \sum_k \delta_k$ , Heathrow is visited by every van

$\sum_i x_{ijk} = y_{ik}$  for all  $j, k$ . if location  $j$  is visited by van  $k$ , then there is exactly one link in

$\sum_i x_{jik} = y_{ik}$  for all  $j, k$ . if location  $j$  is visited by van  $k$ , then there is exactly one link out.

In order to avoid a lot of symmetric solutions (i.e. interchanging identical vans), we can stipulate

$$\sum_i y_{i1} \geq \sum_i y_{i2} \geq \cdots \geq \sum_i y_{i6}.$$

This model has 290 constraints and 1356 variables (all integer).

This is a *relaxation* of the true model as the subtour elimination constraints have been ignored. It will, almost certainly be necessary to append some of them, possibly automatically during the course of optimization.

Having determined the minimum number of vans needed, we fix the values of the  $\delta_k$  variables, introduce slack variables into the time limit constraints and minimize the maximum of these.

## 13.28 Protein folding

We define the following binary variables

$x_{ij} = 1$ , iff hydrophobic acid  $i$  is matched with acid  $j$  ( $i < j$ ). (This does not include those matchings that are predefined by virtue of the acids being contiguous in the chain.)

$y_i = 1$ , iff a fold occurs between the  $i$ th and  $(i + 1)$ st acids in the chain.

For each pair of hydrophobic acids  $i$  and  $j$ , we can match them if (i) they are not contiguous, that is, already matched, (ii) they have an even number of acids between them in the chain and (iii) there is exactly one fold between  $i$  and  $j$ . This gives rise to the constraints